

Research Note

Does NIFTY 50 Recover After a Significant One-Day Fall? An Event Study, 2007–2026

Prepared for: AlgoChowk — Quant Engineer Intern Challenge | Author: Sumit Yadav | Date: 22 Sep 2026 | Repo: github.com/sumityadav026/Algochowk-Nifty-Research

1. Hypothesis

H1: Following a day on which the NIFTY 50 closes down 2.0% or more from the previous close (“event day”), the average return over the subsequent 5 trading days exceeds the average 5-trading-day return earned on non-event days over the same period.

H0 (null): 5-day forward returns following an event are statistically indistinguishable from 5-day forward returns following a random (non-event) day drawn from the same period.

“Recovery” is defined relative to the market’s normal drift, not relative to zero — NIFTY trends upward over multi-year horizons, so a positive post-event return alone would not indicate a genuine effect.

2. Event & Recovery Definitions

Event (“significant fall”)

- **Primary rule:** close-to-close return $\leq -2.0\%$ (fixed threshold).
- **Rationale:** NIFTY’s typical daily move is roughly $\pm 1\%$, so -2% is a clear outlier without being so rare that too few events remain for inference. Fixed and pre-registered before any results were examined.
- **Robustness alternatives (not used to select the headline result):** -1.5% and -3.0% thresholds; an expanding 5th-percentile-of-returns rule; a 2σ -below-60-day-volatility rule.

Recovery

- Excess 5-day forward return = mean(event-day returns) – mean(non-event-day returns), both computed with an identical entry/exit rule, in the same window.
- Supporting evidence: bootstrap 95% CI of the excess, a random-day Monte-Carlo test (H0: event days behave like any random n days), win rate vs. baseline win rate.

Independence of events

- Falls cluster in time (e.g., Oct 2008, Mar 2020). An event is discarded if it occurs within one holding period of the prior accepted event, so trades do not overlap and are treated as independent observations.

3. Entry / Exit

- **Entry:** Open of the trading day immediately following the event day. The -2% return is only observable at the close of the event day, so entering at that close would assume information not yet available (look-ahead bias).
- **Exit:** Close of (event day + H) trading days, H fixed in advance (Section 4).
- **Position sizing:** one unit of capital per trade, one position open at a time (used identically for the baseline).

4. Holding Period

- Primary: H = 5 trading days ($\approx$ one trading week), fixed before results were examined.
- Robustness grid: $H \in \{1, 3, 5, 10\}$ trading days, crossed with threshold and entry-rule variants (26 configurations total).

5. Test Period

- Full sample: 2007-09-17 to 2026-09-18 (4,663 trading days after cleaning).
- Development (in-sample): 2007-09-17 to 2018-12-31 — used for all design and robustness work.
- Out-of-sample: 2019-01-01 to 2026-09-18 — evaluated exactly once, with the Section 2–4 parameters already locked. Trades whose exit date crosses the split boundary are excluded from both windows to prevent leakage.

6. Transaction Costs

- 3 bps commission/statutory charges + 2 bps slippage, per side $\rightarrow$ 10 bps (0.10%) round trip, applied to every simulated trade.
- Sensitivity tested from 0 bps to 80 bps round trip to establish how much of any apparent edge is cost-fragile (Section 7).

7. Key Assumptions

- Trades execute at the printed Open/Close with no partial fills or capacity limits; the index is tradable via a futures/ETF proxy at the assumed cost.
- Price-return index (dividends excluded) — immaterial for a 1–10 day holding horizon.
- All signals use only information available at the time of the decision (verified by unit test).
- Data: NIFTY 50 daily OHLC, Yahoo Finance ($\wedge$ NSEI). Cleaned for duplicates, ordering, invalid OHLC and calendar gaps; three of the largest single-day falls (2020-03-23, 2008-10-24, 2008-01-21) were independently cross-checked against a second source and matched.

8. Headline Result (development, locked configuration: -2.0%, H=5, next-open entry)

Sample	n	Event mean	Baseline mean	Excess	95% CI (excess)	Random-day p
Development	95	+0.02%	+0.11%	-0.09%	[-1.03%, +0.81%]	0.62
Out-of-sample	36	+0.00%	+0.13%	-0.13%	[-1.84%, +1.31%]	0.65

No excess return survives beyond what normal drift would produce, in either window; both confidence intervals contain zero and both random-day tests are far from conventional significance. Of 26 robustness configurations, 0 remained significant after Holm correction; the sign of the excess return was unstable across thresholds and holding periods. Excluding the 2008-09 and 2020 crisis clusters flips the sign to marginally positive (+0.10%) but the CI still contains zero (n=92). A simple net-of-cost backtest returned -22.7% over the full sample versus +419% for buy-and-hold, with a -52.6% maximum drawdown.

9. Conclusion

Hypothesis → Method → Evidence → Baseline → Robustness → Out-of-Sample → Limitations → Conclusion:

The data do not support H1. Post-fall 5-day returns (mean +0.02% development, +0.00% out-of-sample) are statistically indistinguishable from the market's normal 5-day drift (+0.11%, +0.13%); both 95% confidence intervals on the excess contain zero, and both random-day Monte-Carlo tests return $p \approx 0.6-0.65$. The finding is stable across the chronological out-of-sample split, is not an artifact of a single crash cluster (removing 2008-09/2020 flips the sign only marginally, still within a zero-including CI), and is not a product of favorable parameter selection (0 of 26 robustness configurations survive Holm correction). A net-of-cost backtest confirms the result is not economically exploitable: -22.7% versus +419% for buy-and-hold, with a -52.6% drawdown.

The honest conclusion is “no evidence of a reliable recovery effect,” not “proof no effect exists” — with ~95 development events the design has statistical power to detect only an excess of roughly 1.1% or larger, so a smaller true effect cannot be ruled out. What would change this conclusion: a positive, zero-excluding excess CI that persists out-of-sample and across neighbouring thresholds/holding periods, and that survives realistic transaction costs. Full evidence, the 26-cell robustness grid, out-of-sample detail, and independent cross-checks of the code and data are documented in README.md and results/summary.md in the accompanying repository.